

Introduction to Computer and Computer Science

Homework Assignment #4

Due: 10/21 0:00 am

In this homework assignment, you will practice how to apply your own functions you had created in last homework to solve one of the most difficult issues in your entire life – managing your money for something in limited time. As you have learned all the related knowledge during the lectures (exhaustive enumeration, bisection search and so on...), you should have the ability to accomplish this.

Problem #1: Finding the right way

Last time you had explored how both the percentage of your salary that you save each month and your annual raise affect how long it takes you to save for the down payment. However, in reality, you might want to set a particular goal, e.g. to be able to afford the down payment in four years. How much should you save each month to achieve this? In this problem, you are going to write a program to solve this issue.

Before you start to write the code, here are some basic assumptions for this problem:

1. Call the cost of your target **target_price**.
2. Assuming the down payment is 25% of your target price, which leads you this result **down_payment** = $0.25 \times \text{target_price}$.
3. Assuming your monthly salary is **monthly_salary**.
4. Assuming you are going to dedicate a certain percentage of your monthly salary for the down payment. Call it **saving_rate**. Be careful, this number should be in decimal form, i.e. **saving_rate** = 0.87 for 87%.
5. Call the amount that you have saved **current_savings**. You should start with **current_savings** = 0 unless you have wonderful parents or you have “Fa Đà Cái”.
6. Assuming you also invest your current savings wisely, with an annual return of **r**. In other words, your account of current savings will receive an additional money **current_savings** $\times$ (**r**/12) at the end of each month. For simplicity, let assume **r** = 0.04.
7. Call your semi-annual salary raise **semi_annual_raise**. Again, it should be a decimal number, i.e., **semi_annual_raise** = 0.05 for 5%.
8. Raise your salary by this rate in every six months.
9. Call your objective as **target_month**. For simplicity, let us assume the value of this variable is an integer.

Please write function called “**findSavingRate**” that aims to find the best saving rate for your down payment in certain months. The parameters of this function **findSavingRate** should be:

findSavingRate(target_price, monthly_salary, semi_annual_raise, target_month)

Please test your function **findSavingRate** by asking users to enter this information:

1. Her/his monthly salary
2. The price of her/his target
3. The semi-annual salary raise
4. How many months does she/he expect?

You can test your function with these cases:

Case 1

Please enter your monthly salary: 60000

How much is your target price: 1000000

How much is your semi-annual salary raise: 0.05

How many months do you expect for saving: 36

The best saving rate is: 0.095703125

Case 2

Please enter your monthly salary: 100000

How much is your target price: 94870000

How much is your semi-annual salary raise: 0.05

How many months do you expect for saving: 144

The best saving rate is: 0.72265625

Case 3

Please enter your monthly salary: 28000

How much is your target price: 20000000

How much is your semi-annual salary raise: 0.03

How many months do you expect for saving: 36

It is impossible to save enough money for the down payment in 36 months.

Please be aware, the answer of the best saving rate will be slightly different due to different implementations of searching algorithms. An approximate answer (i.e., a close answer) is good enough, you don't need to write a program that prints out a number with exactly same as mine.

Hand in procedure:

As we had mentioned in the lecture, you should list all of your collaborators in your programs. Here is the template:

```

6
7 # =====
8 #
9 # Homework assignment #0
10 # Student: Your name, 你的名字
11 # Student ID: Your student ID, 你的學號
12 # Collaborators: John Doe, Jane Doe, ...
13 #
14 # =====
15 # Your code goes here...
16
17 |

```

Please save your code as a “.py” file, where the file name should follow this format:

108A_hw#?_ID.py

For example,

108A_hw#1_0180506.py

Please be aware. **We are not going to accept any homework file with wrong file name or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the “New e3” system. There will be a section for uploading your homework.